

Local convergence of Newton's method

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be sufficiently smooth, and consider Newton's iteration

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

Let x^* be a root of f , i.e.,

$$f(x^*) = 0.$$

1. Assume that x^* is a simple root:

$$f'(x^*) \neq 0.$$

Prove that there exists a neighborhood of x^* such that, for every initial guess x_0 in this neighborhood, Newton's iteration is well-defined and converges to x^* .

2. Let

$$e_k = x_k - x^*.$$

Assuming $f \in C^2$ in a neighborhood of x^* , show that

$$e_{k+1} = \frac{f''(x^*)}{2f'(x^*)} e_k^2 + o(e_k^2).$$

Deduce that Newton's method converges at least quadratically to x^* . Under what additional condition is the convergence exactly quadratic?

3. Now suppose that x^* is a root of multiplicity $m > 1$, that is,

$$f(x^*) = f'(x^*) = \dots = f^{(m-1)}(x^*) = 0, \quad f^{(m)}(x^*) \neq 0.$$

Assume that f is sufficiently smooth in a neighborhood of x^* .

Show that f can be written locally in the form

$$f(x) = (x - x^*)^m g(x),$$

where g is continuous near x^* and

$$g(x^*) \neq 0.$$

Determine $g(x^*)$ in terms of $f^{(m)}(x^*)$.

Using this representation, show that the standard Newton iteration satisfies

$$\lim_{k \rightarrow \infty} \frac{e_{k+1}}{e_k} = \frac{m-1}{m}, \quad e_k = x_k - x^*.$$

What does this imply about the order of convergence of Newton's method at a multiple root?

4. Consider instead the modified Newton iteration

$$x_{k+1} = x_k - m \frac{f(x_k)}{f'(x_k)}.$$

Prove that this iteration restores quadratic convergence to x^* .

5. Suppose now that the multiplicity m of the root x^* is unknown. Consider the iteration

$$x_{k+1} = x_k - \frac{f(x_k)f'(x_k)}{[f'(x_k)]^2 - f(x_k)f''(x_k)}.$$

Show that this iteration converges quadratically to x^* , provided x_0 is sufficiently close to x^* . *Hint:* Apply Newton's method to

$$F(x) = \frac{f(x)}{f'(x)}.$$